

Edison Meals



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<https://github.com/Hal-9k1>



#20198

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EDUCATION

B.S. Bioengineering: Bioinformatics

University of California: San Diego

- Expected graduation 06/2029
- Junior standing

A.S. Mathematics

College of Alameda

- Graduated 05/2025
- GPA 3.81

LANGUAGES

- C, C++
- JavaScript
- C#
- Lua
- Python
- Java
- TypeScript
- Bash/Zsh/PS

EXPERIENCE

- Applications and systems dev.
- Real-time computing & robotics
- 3D modeling & printing
- MCUs: ESP32, RP2040
- QT GUI
- Unit testing frameworks
- Build systems
- Electron apps & Webpack
- Android app development
- npm ecosystem
- React frontend
- Web development
- Conversational Chinese (Mandarin)

Objective

To obtain an internship gaining practical experience applying programming skills to impactful problems.

Summary

Self-motivated and enthusiastic freshman at UC San Diego studying Bioinformatics with a passion for computer programming, physics, and biology. Looking to explore the unknown at the intersection of biology and computer science.

Projects

Pioneers in Engineering: Dawn v4 | TypeScript

<https://github.com/pioneers/new-dawn>

Led the rewrite of the program used by students in the Pioneers in Engineering high school robotics competition to upload code to and test their robots. Communicates with robot runtimes using SSH and protobuf over TCP. Frontend uses React on Electron and is built with Webpack.

PicoTFT | C++

<https://github.com/Hal-9k1/picotft>

Graphics library written for the RP2040 to control ILI9486 TFT displays. Implements depth buffering, multi-core tiled rendering, Vulkan-like command buffer system, and DMA display communication. Used for various personal projects.

PiE SP25 autonomous | Python, C

<https://github.com/Hal-9k1/pie-spring-2025>

Autonomous program entered in the Pioneers in Engineering 2025 Spring competition. Modular graph-based control system. Features include dynamic-window path selection, localization by template-matching distance sensor output, hot path accelerators written in C that are compiled on-the-fly, and a Python application/resource bundler to comply with the competition's one-file rule. Won the 2025 PiE Software Award.